

CJ products on Today's Deals — strategy & design

2026-07-16 · design approved, implementation pending · companion to `doc/deals-wave-2026-07-16.pdf` · target module `litemall-goods-management` (+ additive `litemall-db`) · migration slot **V40** (V39 taken by the affiliate wave — re-check `flyway_schema_history` before first boot)

Why Today's Deals is local-only today

Barrier	Root cause (verified in code)
No organic CJ discounts	<code>CjProductToNativeAdapter</code> hardcodes <code>counter_price = retail_price</code> for every promoted CJ goods, so <code>discount_pct</code> indexes 0 and <code>deal_flag</code> never fires. CJ's detail API actually returns a <code>suggestSellPrice</code> (MSRP-style anchor) — <code>CjDetailEnrichmentService</code> fetches it and discards it.
Flash deals refuse CJ (errno 652)	Deliberate Phase-B guard: the nightly promote (03:00, <code>promoteBatch</code> over the whole snapshot) and every enrichment (03:30 batch + on-view) rewrite <code>retail_price/counter_price/SKU price</code> unconditionally through <code>CjProductPromotionService.promoteOne</code> . A live price swap would be clobbered within 24 h — or minutes, if someone views the product.

Pricing economics (verified)

- Our CJ retail = `CJ sellPrice` × `usdToCny` (7.2) × **margin (2.0)** — configured in `CJDropshippingConfig.Pricing`, applied in `CjSnapshotSyncService.retailPrice` / `CjDetailEnrichmentService.retail`. There is real margin headroom: deals down to −50% stay at or above cost.
- The CJ order replay (`createOrderV2`) sends **vid + quantity only — no price**. CJ bills wholesale regardless of our charge, so a CJ deal spends only our margin, never CJ's.

Charge-integrity bug discovered during this investigation (Phase-B latent defect — fix first).

Every checkout amount is driven by `litemall_goods_product.price` (the SKU row): cart-add snapshots it (`LitemallOrderOrchestratorService.java:716`) and order-submit authoritatively re-stamps it (`LitemallOrderDomainService.java:53-54`) before the money math. The Phase-B flash-deal swap only touches `litemall_goods.retail_price` — so live deals *display* the deal price (cards, countdown, headline) but **charge the full SKU price at checkout**, and the detail-page SKU buy-box shows the un-swapped price. The fix (Part 1) swaps SKU rows; **no order-service change is needed** — because submit re-reads live SKU prices, charges self-correct the moment the swap lands, and again the moment it unwinds.

Approved strategy — two complementary mechanisms + floor

Decision	Choice (user-approved)
CJ deal sources	Both: organic suggest-price anchors (standing breadth, zero admin work) + curated CJ flash deals (time-boxed countdown offers the admin authors).
Deal floor	Floor at cost: $\text{dealPrice} \geq \text{cjCost} \times \text{litmall.deals.cj-min-margin}$, default 1.0 (never below CJ wholesale). Config can be raised for guaranteed margin or lowered for deliberate loss-leaders.

Part 1 — SKU-level price swap (fixes local charge bug; foundation for CJ)

- **V40:** `litmall_seckill.original_sku_prices` TEXT — JSON `{productId: {"o": original, "s": swapped}}` — plus `litmall_cj_product.suggest_price decimal(10,2)` (Part 3). Hand-edit entities + mappers + `FlywayMigrationTest` (V38 precedent).
- **Activate** (`FlashDealLifecycleTask` stays the only writer): `factor = dealPrice / goods.retailPrice`; every SKU row gets `price = round(max(0.01, original × factor), 2)`, originals captured in the JSON. Single-SKU goods land exactly on the deal price; multi-SKU goods get a proportional discount.
- **Unwind:** restore each SKU only if its current price still equals the swapped value (per-SKU admin-wins, mirroring the existing goods-level guard). For `source='cj'`, additionally re-promote from the snapshot after restore — correct even if CJ repriced mid-deal.
- Cart semantics come out right for free: add-during-deal charges the deal; submit-after-expiry re-stamps the restored price.

Part 2 — enable CJ flash deals

- **Floor validation replaces the blanket 652** in `FlashDealService.create/update`: `cost basis = snapshot price ÷ pricing.margin`; `dealPrice ≥ cost × cj-min-margin` or `errno 650` with the computed floor in the message. 652 remains only when no snapshot row exists (cost unknowable).
- **Sync guards at the single choke point** (`CjProductPromotionService`): load the live-swap set once per run (`SELECT goods_id FROM litmall_seckill WHERE price_swapped=1`); for those goods, null the price fields on the goods patch and skip SKU `price` in `upsertProducts` — stock/title/gallery/variants keep syncing. This covers all three write triggers (nightly `promoteBatch`, nightly `enrichment→promote`, on-demand `enrichment→promote`) because all funnel through `promoteOne`.
- Known accepted edge: a NEW CJ variant appearing mid-deal inserts at the normal price (documented in the ADR).

Part 3 — organic CJ anchors from suggestSellPrice

- Enrichment persists `suggest_price = lowerBound(suggestSellPrice) × usdToCny` — **no margin factor** (it is already a retail suggestion; applying margin would double-mark the anchor).
- Adapter: `counter_price = suggest_price` when it exceeds retail, else retail (unchanged).
- Effect: CJ items whose CJ-suggested retail beats our price by $\geq \text{deal-min-pct}$ (10) organically index `deal_flag=1` and appear on /deals with an honest "was" anchor. Selectivity is natural — only items where suggest > 2×cost anchor at all. Coverage grows with enrichment (nightly quota-paced batch + instant on-view).

Part 4 — surfaces & docs

- No customer SPA work: CJ goods are native ids — /deals, cards, countdown chip, DealBanner all already work.
- Admin `DealForm.tsx`: un-grey CJ goods in the picker if blocked; surface the floor error inline.
- Update `adr-flash-deals-price-swap.md` (SKU swap, CJ floor, sync guard, suggest anchor) and `handoff-gateway-admin-flash-deals.md` (652 → missing-snapshot only; 650 floor message).

Verification protocol (when implemented)

1. Compile + `FlywayMigrationTest`; V40 applies (re-check history for a V40 collision first — V39 was taken mid-wave by the affiliate branch).
2. **Charge fix (local)**: deal on goods 1009012 → SKU row 59→41.30 in DB → cart-add through the *running* order service stores 41.30 → submit persists `order_goods.price=41.30` → disable → restored → fresh submit charges 59.
3. **CJ deal**: below-floor price → 650 naming the floor; valid deal → SKUs swap; view the product (on-demand enrichment) and run the promote path → price survives (guard proof); disable → unwind → snapshot-derived price back.
4. **Organic**: `cj-enrich-one` a pid → `suggest_price` lands → promote → counter > retail → reindex → `discount_pct > 0`, item on `/srv/search?deal_flag=1`; report coverage counts.
5. §23 relevance harness at pinned baseline (no scoring change); regression: non-deal CJ prices byte-identical across a promoteBatch, local deals, suggest, facet rail; DealForm accepts a CJ goods; tsc + prod bundles.

Effort shape: one migration (V40), ~5 backend files touched in goods-management (`FlashDealLifecycleTask`, `FlashDealService`, `CjDetailEnrichmentService`, `CjProductToNativeAdapter`, `CjProductPromotionService`) + litemall-db hand-edits + one admin SPA form tweak. Zero order-service changes. Post-merge: rebuild/restart main :8082 goods-management only.